

Slopes and local change

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Loss functions

Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Forward passes by hand

Without opening the old lesson, explain one key idea from Forward passes by hand. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Bias terms and decision boundaries

Without opening the old lesson, explain one key idea from Bias terms and decision boundaries. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Check a symmetric estimate for Slopes and local change. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Slopes and local change. Proposed explanation: “Divide by the step only”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Estimate a new curve independently

Let $J(a) = (a + 1)^2$. At $a = 1$ use step 0.25 to find the symmetric slope. Compare $J(1)$ with $J(0.9)$, then explain why moving to -4 does not follow safely from the slope clue.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Measure an interval slope

For $L(w) = (w - 2)^2$, calculate the average slope from $w = 0$ to $w = 2$. Compare it with the local slope at $w = 2$.

My reasoning and evidence

6. W2 Check a symmetric estimate

At centre $w = 0$, use step 0.5. Calculate $L(-0.5)$, $L(0.5)$ and the symmetric slope.

My reasoning and evidence

7. W3 Challenge a large step

Starting at $w = 3$, compare loss after moving to 2.9 and after moving to -1. Both moves decrease w .

My reasoning and evidence

8. W4 Estimate a new curve independently

Let $J(a) = (a + 1)$ squared. At $a = 1$ use step 0.25 to find the symmetric slope. Compare $J(1)$ with $J(0.9)$, then explain why moving to -4 does not follow safely from the slope clue.

My reasoning and evidence

9. W5 Limit zero slope evidence

For $F(w) = (w^2 - 1)^2$, compare $F(0)$ and $F(0.1)$. Does zero local slope at zero prove a minimum?

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled slopes and local change evidence sheet to Image-pattern Network Investigation. Show how you can calculate an average rate of change over a stated interval. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Slopes and local change

Use a local slope to propose a small move rather than claiming every move helps.

Toy loss $J(a)=(a-2)^2$. At $a = 0$, use symmetric step 0.5.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Slopes and local change

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Evaluate nearby points

$J(0.5)=2.25$ and $J(-0.5)=6.25$.



2. Estimate slope

$\text{Slope} = (2.25 - 6.25) / (0.5 - (-0.5)) = -4$.



3. Check small move

Move to $a = 0.25$: loss 3.0625, below starting loss 4.

CHANGE ONE THING

Move from 0 all the way to 5.

Explain the changed result using the matching lesson.

Which calculation disproves “any increase helps”? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Use a local slope to propose a small move rather than claiming every move helps.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Move to $a = 0.25$: loss 3.0625, below starting loss 4.

1. Check small move
2. Estimate slope
3. Evaluate nearby points

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Check small move Move to $a = 0.25$: loss 3.0625, below starting loss 4. A slope describes how the loss changes near the current parameter. The symmetric calculation compares equal small moves on either side. Its negative result suggests a small increase may help at zero. It does not promise every increase helps: jumping to five passes the optimum at two and produces a larger loss. Check the new loss after each proposed step.

Practice 2: explain the deciding evidence

Changed-case practice: Move from 0 all the way to 5. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. Loss becomes 9, exceeding the starting 4.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Loss becomes 9, exceeding the starting 4. A negative local slope suggests increasing a locally, not that every positive jump decreases loss.

Repair a missing building block

Review Loss functions (ai-5/8). Goal: Calculate per-example squared error and mean squared error. Short practice: Why must the report name the chosen loss? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Forward passes by hand (ai-5/6). Goal: Record scores and activations in dependency order. Short practice: Why should feedback identify h before correcting only y ? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Bias terms and decision boundaries (ai-5/3). Goal: Calculate a weighted score with a bias term. Short practice: Is B positive under the original or changed model? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
